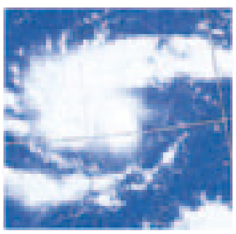
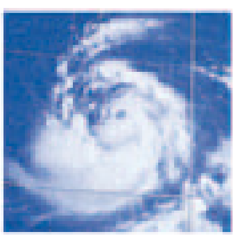
Waves and weather

DOWN THE SPOUT
Water spouts (spinning sprays sucked up from the surface) begin when whirling air drops down from a storm cloud to the ocean.

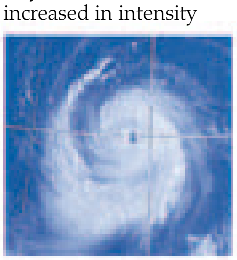
SEAWATER IS CONSTANTLY moving. At the surface, wind-driven waves can be 50 ft (15 m) from crest to trough. Major surface currents are driven by the prevailing winds. Both surface and deep-water currents help modify the world's climate by taking cold water from the polar regions toward the tropics, and vice versa. Shifts in this flow affect life in the ocean. In an El Niño climatic event, warm water starts to flow down the west of South America, which stops nutrient-rich, cold water rising up, causing plankton growth to slow and fisheries to fail. Heat from oceans creates air movement, from swirling hurricanes to daytime breezes on-shore, or nighttime ones off-shore. Breezes occur as the ocean heats up more slowly than the land in the day. Cool air above the water blows in, replacing warm air above the land, and the reverse at night.



Day 2: Thunderstorms as swirling cloud mass

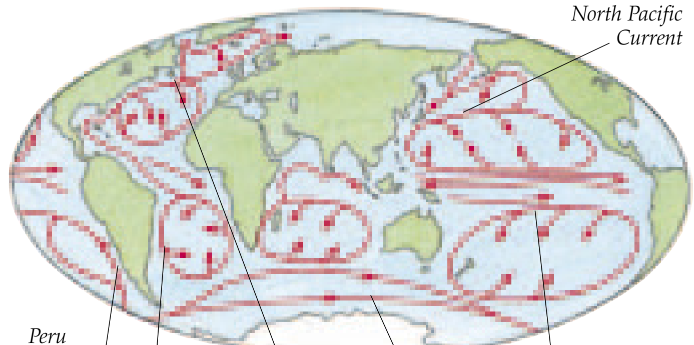


Day 4: Winds have increased in intensity

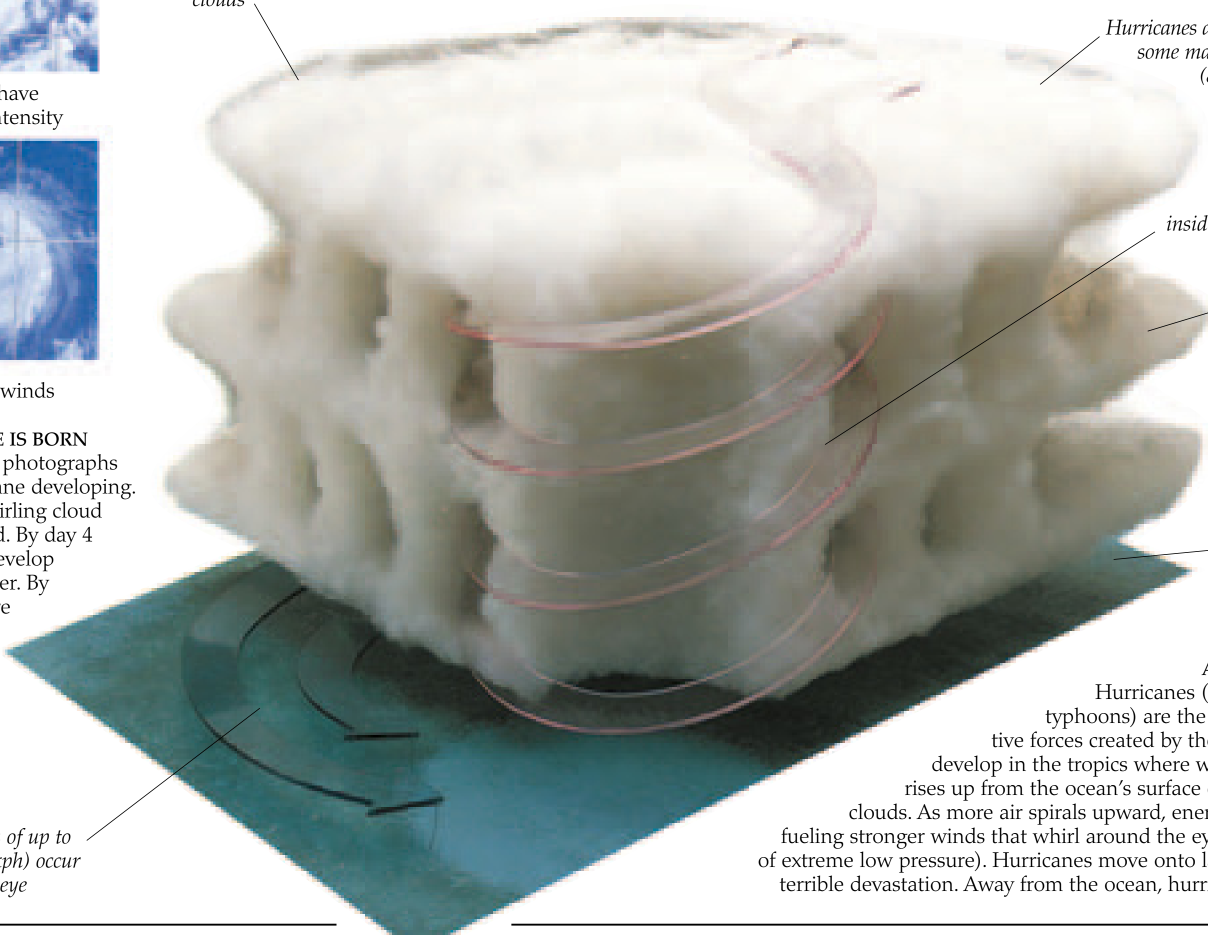


Day 7: Strong winds

A HURRICANE IS BORN
These satellite photographs show a hurricane developing. On day 2 a swirling cloud mass is formed. By day 4 fierce winds develop about the center. By day 7 winds are the strongest.



RIVERS OF THE SEA
Currents are huge masses of water moving through the oceans. The course currents follow is not precisely the same as the trade winds and westerlies, because currents are deflected by land and the Coriolis Force produced by the Earth's rotation. The latter causes currents to shift to the right in the northern hemisphere and to the left in the southern. There are also currents that flow due to differences in density of seawater.



HEART OF A HURRICANE
Hurricanes (also known as typhoons) are the most destructive forces created by the oceans. They develop in the tropics where warm, moist air rises up from the ocean's surface creating storm clouds. As more air spirals upward, energy is released, fueling stronger winds that whirl around the eye (a calm area of extreme low pressure). Hurricanes move onto land and cause terrible devastation. Away from the ocean, hurricanes die out.